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Inventor(s)	Li; Zixuan et al.

Splitting spinning apparatus

Abstract

Disclosed is a splitting spinning apparatus, including: a base, a turntable for supporting a blank, a rotating mechanism for driving the turntable, a mandrel disposed above the turntable, and a first drive mechanism for driving the mandrel to move along a vertical direction, a roller being provided at one side of the turntable, a second drive mechanism and a third drive mechanism being disposed on the base; the splitting spinning apparatus further includes a squeezing roll and a fourth drive mechanism; the turntable includes an inner plate, an outer plate, and a hold mechanism, the blank having a diameter greater than that of the inner plate and less than an outer diameter of the outer plate, an upper side of the inner plate being in flush with an upper side of the outer plate, an outer diameter of the squeezing roll being equal to the diameter of the inner plate.

Inventors: Li; Zixuan (Ningbo, CN), Shu; Xuedao (Ningbo, CN), Leng; Junkun (Ningbo, CN), Niu; Xiaomiao (Ningbo, CN)

Applicant: NINGBO UNIVERSITY (Zhejiang, CN)

Family ID: 87914151

Assignee: NINGBO UNIVERSITY (Ningbo, CN)

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Primary Examiner: Sullivan; Debra M

Attorney, Agent or Firm: Oliff PLC

Background/Summary

FIELD

(1) The disclosure pertains to the technical field of splitting spinning apparatuses, and more particularly relates to a splitting spinning apparatus.

BACKGROUND

(2) Dual-cylinder structures have a wide array of applications, e.g., flame tubes for turbojet and turbofan engines. The dual-cylinder structure comprises a bottom plate, an inner cylinder, and an outer cylinder. Conventional splitting spinning apparatuses cannot work the outer cylinder. For example, a splitting spinning apparatus has been proposed in the *Study on Dividing Spinning Process of Cylindrical Parts With Bottom Flange* published in the journal "Aeronautical Manufacturing Technology," where the splitting spinning apparatus comprises a backing plate, a mandrel, and a roller; upon working, the mandrel compresses a blank tightly on the backing plate, then the roller moves along the radial direction of the blank to splitting spin the upper side of the blank, whereby an edge of the blank is bifurcated into a baseplate and a big-end-up trumpet flange; after the roller gets close to the mandrel, the roller moves upward to spin the trumpet flank, where the trumpet flange is deformed to form an inner cylinder fitted to the outer side of the mandrel. However, existing splitting spinning apparatuses cannot continue spinning the baseplate to work an outer cylinder.

SUMMARY

(3) To address a drawback of conventional splitting spinning apparatuses which cannot work an outer cylinder, the disclosure provides a splitting spinning apparatus which can work an outer cylinder.

(4) To achieve the objective above, the disclosure adopts a technical solution summarized below: A splitting spinning apparatus, comprising a base, a turntable for supporting a blank, a rotating

mechanism for driving the turntable, a mandrel disposed above the turntable, and a first drive mechanism for driving the mandrel to move along a vertical direction, a roller being provided at one side of the turntable, a second drive mechanism for driving the roller to move along a radial direction of the blank and a third drive mechanism for driving the roller to move along the vertical direction being disposed on the base; wherein the splitting spinning apparatus further comprises a squeezing roll disposed above the turntable and a fourth drive mechanism for driving the squeezing roll to move along the vertical direction, the squeezing roll and the mandrel being coaxially arranged; wherein the turntable comprises an inner plate rotatably connected to the base, an outer plate sleeved outside of the inner plate and slidable along an axial direction of the inner plate, and a hold mechanism for holding the outer plate in place, the blank having a diameter greater than a diameter of the inner plate and less than an outer diameter of the outer plate, an upper side of the inner plate being in flush with an upper side of the outer plate, an outer diameter of the squeezing roll being equal to the diameter of the inner plate.

(5) With the above arrangement, the roller first splitting spins the blank to cause an edge of the blank to be bifurcated into a baseplate and a trumpet flange; then, the roller moves upward to spin the trumpet flange to form the inner cylinder; for continuously working the outer cylinder, the outer plate moves downward to expose the lower side of the baseplate to the air; after the squeezing roll compresses the baseplate tightly against the inner plate, the roller moves along the radial direction of the blank, and then the edge of the baseplate tilts upward under the action of the roller, such that after the roller gets close to the squeezing roll, the roller moves upward to cause the baseplate to be continuously deformed to form the outer cylinder fitted to the outer wall of the squeezing roll.

(6) Furthermore, the roller comprises a first state and a second state; when the roller is in the first state, one end of the roller proximal to the turntable tilts downward for the roller to splitting spin the upper side of the blank to form an inner cylinder; when the roller is in the second state, one end of the roller proximal to the turntable tilts upward, such that the roller spins the baseplate to form an outer cylinder.

(7) With the arrangement above, the disclosure eliminates the need of first processing the edge of the baseplate when working the outer cylinder; instead, the roller in the second state moves directly upward to spin the baseplate to form the outer cylinder, whereby the disclosure improves the efficiency of working the outer cylinder.

(8) Furthermore, a lifting platform is disposed at one side of the turntable; the third drive mechanism is disposed on the base and connected to the lifting platform, an upper side of the lifting platform is slidably connected to a sliding seat; the second drive mechanism is disposed on the lifting platform and connected to the sliding seat; a roller carrier is provided at one side of the sliding seat proximal to the turntable, one end of the roller carrier proximal to the turntable being rotatably connected to the roller, the roller carrier being rotatably connected to the sliding seat, whereby state switching of the roller is enabled by turning the roller carrier; and the sliding seat is provided with a locking mechanism for locking the roller carrier.

(9) The above arrangement facilitates switching the states of the roller, further enhancing production efficiency of the disclosure.

(10) Furthermore, the roller carrier comprises a rotating seat rotatably connected to the sliding seat and a bracket securely connected to one side of the rotating seat proximal to the turntable; the roller is rotatably connected to one end of the bracket proximal to the turntable; and the locking mechanism comprises a retaining ring securely connected to the sliding seat and a locking bolt thread-fitted to the retaining ring, the rotating seat being rotatably connected inside the retaining ring, the locking bolt abutting against the rotating seat.

(11) The arrangement above facilitates tightly locking the roller carrier.

(12) Furthermore, a first limiting rod is securely connected to an upper side of the retaining ring, a second limiting rod is securely connected to a lower side of the retaining ring, and a rotating rod is securely connected to the rotating seat; wherein when the rotating rod abuts against the first

limiting rod, the roller is disposed in the first state; when the rotating rod abuts against the second limiting rod, the roller is disposed in the second state.

(13) With the arrangement above, positioning of the roller carrier is facilitated when the roller is switched between the states by turning the roller carrier.

(14) Furthermore, a top plate is disposed above the base, the top plate being securely connected to the base via a connecting plate, and the first drive mechanism is configured as a first oil cylinder, the first oil cylinder comprising a first cylinder body securely connected to a lower side of the top plate, a first piston slidably connected inside the first cylinder body, a first piston rod securely connected to a lower side of the first piston, and a first oil pump connected to the first cylinder body, a lower end of the first piston rod being rotatably connected to the squeezing roll.

(15) Furthermore, the fourth drive mechanism comprises a shell sleeved over the first cylinder body, and a first cavity surrounding the first cylinder body is formed between the shell and the first cylinder body; and the fourth drive mechanism further comprises a second piston slidingly connected inside the first cavity, a second piston rod securely connected to a lower side of the second piston, and a second oil pump in communication with the first cavity, a mounting plate being securely connected to a lower end of the second piston rod, the squeezing roll being rotatably connected to a lower side of the mounting plate.

(16) Furthermore, a rotating plate is disposed between the mounting plate and the squeezing roll, the rotating plate and the mounting plate being both of an annular shape, the first piston rod passing through the rotating plate and the mounting plate, the rotating plate and the squeezing roll being detachably connected; a guide ring extending upward is securely connected to an inner side of the rotating plate; a limiting protrusion extending outward is securely connected to an upper side of the guide ring; an inner side of the mounting plate is disposed between the limiting protrusion and the rotating plate; a rolling bearing is disposed between the mounting plate and the guide ring; a thrust bearing is disposed between the limiting protrusion and the mounting plate; and the thrust bearing is further disposed between the mounting plate and the rotating plate.

(17) The arrangement above increases rotating stability of the squeezing roll and also reduces resistance against rotation of the squeezing roll.

(18) Furthermore, the rotating mechanism comprises an output seat rotatably connected to the base, the inner plate being disposed at an upper side of the output seat and securely connected to the output seat via a connecting post; wherein the rotating mechanism further comprises a driven wheel securely coupled to the output seat, an electric motor disposed a lower side of the base, and a driving wheel attached to the electric motor, the driving wheel being in transmission connection with the driven wheel via a belt, a radius of the driving wheel being smaller than a radius of the driven wheel.

(19) The arrangement above amplifies the force outputted from the output seat.

(20) Furthermore, an endplate is securely attached to an upper end of the squeezing roll, the endplate being detachably connected to the rotating plate; the first piston rod passes through the endplate and is in sealed connection with the endplate; an O-shaped ring is disposed between the first piston rod and the endplate; a first gas channel passing through upper and lower sides of the endplate is provided on the endplate, a lower end of the first gas channel extending till an inner side of the squeezing roll, the first gas channel being provided with a one-way valve facing the inner side of the squeezing roll; the hold mechanism comprises a slide bushing sleeved over the connecting post and a holding bolt in threaded connection with the slide bushing, the holding bolt abutting against the connecting post; a plurality of catching grooves are provided at an upper side of the inner plate; a second gas channel is provided inside the connecting post, an upper end of the second gas channel communicating with the catching grooves; a second cavity is provided at an inner side of the output seat, a third piston being slidably attached inside the second cavity, the third piston partitioning the second cavity into an upper chamber disposed at an upper side of the third piston and a lower chamber disposed at a lower side of the third piston, the upper chamber

communicating with the second gas channel, the lower chamber communicating with the air; and the lower chamber is provided with a spring exerting an upward force against the third piston; a connecting rod is securely connected to an upper side of the third piston, and an upper end of the connecting rod abuts against the slide bushing.

(21) With the above arrangement, after the inner cylinder is completely worked and the squeezing roll compresses the baseplate tightly against the inner plate, it is easy to pull the mandrel out of the inner cylinder; to work the outer cylinder, the outer plate moves downward to bring the third piston to move downward, where the third piston draws out the air in the catching grooves such that the catching grooves securely hold the blank; after the outer cylinder is completely worked, the first piston rod moves downward to increase the air pressure in the squeezing roll, whereby the squeezing roll is easily pulled out from the outer cylinder, which improves demolding efficiency of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of an example splitting spinning apparatus.
- (2) FIG. 2 is an enlarged view of part A in FIG. 1.
- (3) FIG. 3 is a schematic diagram of a roller splitting spinning a blank.
- (4) FIG. 4 is a schematic diagram of a roller spinning a trumpet flange to work an inner cylinder.
- (5) FIG. 5 is a schematic diagram of the roller working a baseplate.
- (6) FIG. 6 is a schematic diagram of the roller working an outer cylinder.
- (7) FIG. 7 is a schematic diagram of the roller in a second state moving to the lower side of the baseplate.
- (8) FIG. 8 is a schematic diagram of the roller in the second state working an outer cylinder.
- (9) FIG. 9 is a schematic diagram of extracting a squeezing roll from the outer cylinder.

DETAILED DESCRIPTION

(10) Hereinafter, the technical solution of the disclosure will be further explained through embodiments with reference to the accompanying drawings.

(11) Referring to FIGS. 1 through 9, a splitting spinning apparatus comprises a base **11**, a turntable **12** for supporting a blank **21**, a rotating mechanism **13** for driving the turntable **12**, a mandrel **14** disposed above the turntable **12**, and a first drive mechanism for driving the mandrel **14** to move along a vertical direction, where a roller **16** is provided at one side of the turntable **12**, a second drive mechanism **17** for driving the roller **16** to move along the radial direction of the blank **21** and a third drive mechanism **18** for driving the roller **16** to move along the vertical direction are disposed on the base **11**; the splitting spinning apparatus further comprises a squeezing roll **19** disposed above the turntable **12** and a fourth drive mechanism for driving the squeezing roll **19** to move along the vertical direction, where the squeezing roll **19** and the mandrel **14** are coaxially arranged; the turntable **12** comprises an inner plate **121** rotatably connected to the base **11**, an outer plate **122** sleeved outside of the inner plate **121** and slidable along the axial direction of the inner plate **121**, and a hold mechanism **123** for holding the outer plate **122** in place, where the blank **21** has a diameter greater than that of the inner plate **121** but less than the outer diameter of the outer plate **122**, the upper side of the inner plate **121** is in flush with the upper side of the outer plate **122**, and the outer diameter of the squeezing roll **19** is equal to the diameter of the inner plate **121**.

(12) With the above arrangement, the roller **16** first splitting spins the blank **21** to cause an edge of the blank **21** to be bifurcated into a baseplate **22** and a trumpet flange **23**; then, the roller **16** moves upward to spin the trumpet flange **23** to form the inner cylinder; for continuously working the outer cylinder, the outer plate **122** moves downward to expose the lower side of the baseplate **22** to the air; after the squeezing roll **19** compresses the baseplate **22** tightly against the inner plate **121**, the

roller **16** moves along the radial direction of the blank **21**, and then the edge of the baseplate **22** tilts upward under the action of the roller **16**, such that after the roller **16** gets close to the squeezing roll **19**, the roller **16** moves upward to cause the baseplate **22** to be continuously deformed to form the outer cylinder fitted to the outer wall of the squeezing roll **19**.

(13) Specifically, the roller **16** herein may be a conventional roller. In operation, the inner cylinder is worked before the outer cylinder. When working the inner cylinder, a round blank **21** is first placed on the turntable **12**; the first drive mechanism drives the mandrel **14** to move downward to compress the blank **21** tightly against the turntable **12**; the inner plate **121** and the outer plate **122** serve to support the blank **21**; the rotating mechanism **13** is activated to bring the turntable **12**, the blank **21**, and the mandrel **14** to rotate synchronously, and the second drive mechanism **17** drives the roller **16** to move towards the mandrel **14** along the radial direction of the blank **21**; when the roller **16** engages the edge of the blank **21**, the blank **21** brings the roller **16** to rotate; as the roller **16** moves continuously, the end portion of the roller **16** splitting spins the blank **21**, causing the edge of the blank **21** to be bifurcated to form the baseplate **22** and the trumpet flange **23**, as illustrated in FIG. 3. After the roller **16** gets close to the mandrel **14**, the third drive mechanism **18** drives the roller **16** to move upward to spin the trumpet flange **23**, whereby the trumpet flange **23** is deformed to form the inner cylinder fitted to the outer side of the mandrel **14**, as illustrated in FIG. 4. To work the outer cylinder, the second drive mechanism **17** drives the roller **16** away from the mandrel **14**, the hold mechanism **123** releases the outer plate **122** such that the outer plate **122** moves downward to be separated from the baseplate **22**, causing the lower side of the baseplate **22** exposed in the air; then, the squeezing roll **19** moves downward to compress the baseplate **22** tightly against the inner plate **121**, and the third drive mechanism **18** drives the roller **16** to move downward to cause the lower end of the roller **16** to be lower than the lower side of the baseplate **22**; after the rotating mechanism **13** brings the blank **21**, the squeezing roll **19**, and the turntable **12** to rotate synchronously, the second drive mechanism **17** drives the roller **16** to get close to the baseplate **22**; after the roller **16** engages the baseplate **22**, the baseplate **22** brings the roller **16** to rotate; as the roller **16** proceeds continuously, the edge of the baseplate **22** tilts upward under the action of the roller **16**; when the roller **16** gets close to the squeezing roll **19**, the third drive mechanism **18** drives the roller **16** to move upward to spin the deformed baseplate **22** such that the baseplate **22** continues deformed to form an outer cylinder fitted to the outer wall of the squeezing roll **19**, as illustrated in FIG. 6.

(14) As an implementation manner, the roller **16** comprises a first state and a second state; when the roller **16** is in the first state, one end of the roller **16** proximal to the turntable **12** tilts downward for the roller **16** to splitting spin the upper side of the blank **21** to form the inner cylinder; when the roller **16** is in the second state, one end of the roller **16** proximal to the turntable **12** tilts upward, such that the roller **16** spins the baseplate **22** to form the outer cylinder.

(15) With the arrangement above, the disclosure eliminates the need of first processing the edge of the baseplate **22** when working the outer cylinder; instead, the roller **16** in the second state moves directly upward to spin the baseplate **22** to form the outer cylinder, whereby the disclosure improves the efficiency of working the outer cylinder. In addition, when the roller in the first state works the blank, the upper side of the roller is worn, while when the roller is switched to the second state, the upper side and the lower side of the roller are reversed, such that when the roller in the second state moves upward to spin the baseplate, the opposite lower side of the roller is worn. In view of the above, by switching the roller' states to produce a cylindrical structure, both of the upper and lower sides of the roller can be worn; as a result, uniform wear of the roller further extends its service life.

(16) Specifically, when working the inner cylinder, the roller **16** is in the first state, which facilitates the lower end of the roller **16** to splitting spin the blank **21**; after the edge of the blank **21** is bifurcated, the trumpet flange **23** extends obliquely upward along the roller **16**, as illustrated in FIG. 3. After the roller **16** gets close to the mandrel **14**, the roller **16** moves upward to work the

inner cylinder, as illustrated in FIG. 4. When the roller in the first state is working the inner cylinder, the upper side of the roller is worn. To work the outer cylinder, after the outer plate **122** moves downward and the squeezing roll **19** compresses tightly the baseplate **22**, the roller **16** is switched to the second state, where the upper side and lower side of the roller are reversed; the orientation of the roller **16** is adjusted via the second drive mechanism **17** and the third drive mechanism **18**, whereby the upper end of the roller **16** comes close to the edge of the inner plate **121**, as illustrated in FIG. 7; at this point, after the rotating mechanism **13** drives the blank **21**, the turntable **12**, and the squeezing roll **19** to rotate synchronously, the third drive mechanism **18** drives the roller **16** to move upward and the roller **16** spins the baseplate **22**, causing the baseplate **22** deformed to form the outer cylinder fitted to the outer wall of the squeezing roll **19**, as illustrated in FIG. 8. When the roller in the second state moves upward to spin the baseplate, the opposite side of the roller is worn; in view of the above, by switching between the roller's states to work a cylindrical structure, both of the upper and lower sides of the roller can be worn uniformly, which further extends service life of the roller.

(17) As an implementation manner, a lifting platform **111** is disposed at one side of the turntable **12**, where the third drive mechanism **18** is disposed on the base **11** and connected to the lifting platform **111**; the upper side of the lifting platform **111** is slidably connected to a sliding seat **112**, where the second drive mechanism **17** is disposed on the lifting platform **111** and connected to the sliding seat **112**; a roller carrier **113** is provided at the side of the sliding seat **112** proximal to the turntable **12**, where one end of the roller carrier **113** proximal to the turntable **12** is rotatably connected to the roller **16**, and the roller carrier **113** is rotatably connected to the sliding seat **112**, facilitating switching between the states of the roller **16** by turning the roller carrier **113**. The sliding seat **112** is provided with a locking mechanism for locking the roller carrier **113**.

(18) The above arrangement facilitates switching between the states of the roller **16**, further enhancing production efficiency of the disclosure.

(19) Specifically, when the third drive mechanism **18** drives the lifting platform **111** to move vertically, the lifting platform **111** brings the sliding seat **112**, the roller carrier **113**, and the roller **16** to move upward; when the second drive mechanism **17** drives the sliding seat **112** to move along the lifting platform **111**, the sliding seat **112** brings the roller carrier **113** and the roller **16** to move along the radial direction of the blank **21**. After the locking mechanism is unlocked, the roller carrier **113** may rotate about the horizontal axis, where the rotating range of the roller carrier **113** is 180°; the roller carrier **113** brings the roller **16** to rotate, causing the roller **16** to be in the first state or the second state; after the state switching is complete, the roller carrier **113** is locked tightly by the locking mechanism **113**. Specifically, the second drive mechanism may be provided as an oil cylinder, while the third drive mechanism **18** may also be provided as an oil cylinder.

(20) As an implementation manner, the roller carrier **113** comprises a rotating seat **1131** rotatably connected to the sliding seat **112** and a bracket **1132** securely connected to the side of the rotating seat **1131** proximal to the turntable **12**; the roller **16** is rotatably connected to the end of the bracket **1132** proximal to the turntable **12**; the locking mechanism comprises a retaining ring **1133** securely connected to the sliding seat **112** and a locking bolt **1134** thread-fitted to the retaining ring **1133**, where the rotating seat **1131** is rotatably connected inside the retaining ring **1133**, and the locking bolt **1134** abuts against the rotating seat **1131**.

(21) The arrangement above facilitates locking of the roller carrier **113**.

(22) Specifically, the rotating seat **1131** has a cylindrical shape; the axial line of the rotating seat **1131** extends along the horizontal direction; the retaining ring **1133** enables the rotating seat **1131** to rotate more stably; when the locking bolt **1134** is unscrewed, the locking bolt **1134** is disengaged from the rotating seat **1131**, and the rotating seat **1131** may rotate about the axial line of the rotating seat **1131**, thereby switching between the states of the roller **16**; after the switching is complete, the locking bolt **1134** is screwed tightly, whereby the rotating seat **1131** may be locked tightly.

(23) As an implementation manner, a first limiting rod **1135** is securely connected to the upper side

of the retaining ring **1133**, a second limiting rod **1136** is securely connected to the lower side of the retaining ring **1133**, and a rotating rod **1137** is securely connected to the rotating seat **1131**, where when the rotating rod **1137** abuts against the first limiting rod **1135**, the roller **16** is disposed in the first state; when the rotating rod **1137** abuts against the second limiting rod **1136**, the roller **16** is disposed in the second state.

(24) With the arrangement above, positioning of the roller carrier **113** is facilitated when the roller **16** is switched between the states by turning the roller carrier **113**.

(25) As an implementation manner, a top plate **115** is disposed above the base **11**, the top plate **115** is securely connected to the base **11** via the connecting plate, and the first drive mechanism is configured as a first oil cylinder, where the first oil cylinder comprises a first cylinder body **151** securely connected to the lower side of the top plate **115**, a first piston **152** slidably connected inside the first cylinder body **151**, a first piston rod **153** securely connected to the lower side of the first piston **152**, and a first oil pump connected to the first cylinder body **151**, where the lower end of the first piston rod **153** is rotatably connected to the squeezing roll **19**.

(26) The first oil cylinder is fully filled with oil; when the first oil pump delivers the oil to the upper end of the oil cylinder, the pressure at the upper side of the first piston **152** increases, causing the oil at the lower side of the first piston **152** to be discharged from the first oil cylinder, whereby the first piston **152** brings the first piston rod **153** and the mandrel **14** to move downward; likewise, when the first oil pump delivers the oil to the lower end of the first oil cylinder, the pressure at the lower side of the first piston **152** increases, causing the oil at the upper side of the first piston **152** to be discharged from the first oil cylinder, whereby the first piston **152** brings the first piston rod **153** and the mandrel **14** to move upward.

(27) As an implementation manner, the fourth drive mechanism comprises a shell **1911** sleeved over the first cylinder body **151**, and a first cavity surrounding the first cylinder body **151** is formed between the shell **1911** and the first cylinder body **151**; the fourth drive mechanism further comprises a second piston **1912** slidably connected inside the first cavity, a second piston rod **1913** securely connected to the lower side of the second piston **1912**, and a second oil pump in communication with the first cavity, where a mounting plate **1914** is securely connected to the lower end of the second piston rod **1913** and the squeezing roll **19** is rotatably connected to the lower side of the mounting plate **1914**.

(28) The first cavity is fully filled with oil; when the second oil pump delivers the oil to the upper end of the first cavity, the pressure at the upper side of the second piston **1912** increases, causing the oil at the lower side of the second piston **1912** to be discharged from the first cavity, whereby the second piston **1912** brings the second piston rod **1913**, the mounting plate **1914**, and the squeezing roll **19** to move downward; when the second oil pump delivers the oil to the lower end of the first cavity, the pressure at the lower side of the second piston **1912** increases, causing the oil at the upper side of the second piston **1912** to be discharged from the first cavity, whereby the second piston **1912** brings the second piston rod **1913**, the mounting plate **1914**, and the squeezing roll **19** to move upward.

(29) As an implementation manner, a rotating plate **1915** is disposed between the mounting plate **1914** and the squeezing roll **19**; the rotating plate **1915** and the mounting plate **1914** are both of an annular shape; the first piston rod **153** passes through the rotating plate **1915** and the mounting plate **1914**; the rotating plate **1915** and the squeezing roll **19** are detachably connected; a guide ring **19151** extending upward is securely connected to the inner side of the rotating plate **1915**; a limiting protrusion **19152** extending outward is securely connected to the upper side of the guide ring **19151**; the inner side of the mounting plate **1914** is disposed between the limiting protrusion **19152** and the rotating plate **1915**; a rolling bearing **19153** is disposed between the mounting plate **1914** and the guide ring **19151**; a thrust bearing **19154** is disposed between the limiting protrusion **19152** and the mounting plate **1914**; and the thrust bearing **19154** is also disposed between the mounting plate **1914** and the rotating plate **1915**.

(30) The arrangement above increases rotating stability of the squeezing roll **19** and also reduces resistance against rotation of the squeezing roll **19**.

(31) Specifically, when the inner side of the mounting plate **1914** is disposed between the limiting protrusion **19152** and the rotating plate **1915**, vertical play of the squeezing roll **19** is prevented, where the inner side of the rolling bearing **19153** is fitted to the guide ring **19151** and the external side of the rolling bearing **19153** is fitted to the mounting plate **1914**, whereby transverse play of the squeezing roll **19** is prevented.

(32) As an implementation manner, the rotating mechanism **13** comprises an output seat **131** rotatably connected to the base **11**, where the inner plate **121** is disposed at the upper side of the output seat **131** and securely connected to the output seat **131** via a connecting post **132**; the rotating mechanism **13** further comprises a driven wheel **133** securely coupled to the output seat **131**, an electric motor **134** disposed at the lower side of the base **11**, and a driving wheel attached to the electric motor **134**, where the driving wheel **135** is in transmission connection with the driven wheel **133** via a belt **136**, the radius of the driving wheel **135** being smaller than that of the driven wheel **133**.

(33) With the arrangement above, the force outputted from the output seat **131** is amplified.

(34) Specifically, the electric motor **134** actuates the turntable **12** to rotate via the driving wheel **135**, the belt **136**, the driven wheel **133**, and the output seat **131**. The radius of the driving wheel **135** is smaller than that of the driven wheel **133**, whereby the force outputted from the electric motor **134** may be amplified.

(35) As an implementation manner, an endplate **192** is securely attached to the upper end of the squeezing roll **19**, the endplate **1915** being detachably connected to the rotating plate **1915**; the first piston rod **153** passes through the endplate **192** and is in sealed connection with the endplate **192**; an O-shaped ring **193** is disposed between the first piston rod **153** and the endplate **192**; a first gas channel **1921** passing through the upper and lower sides of the endplate **192** is provided on the endplate **192**; the lower end of the first gas channel **1921** extends till the inner side of the squeezing roll **19**; the first gas channel **1921** is provided with a one-way valve **1922** facing the inner side of the squeezing roll **19**; the hold mechanism **123** comprises a slide bushing **1231** sleeved over the connecting post **132** and a holding bolt **1232** in threaded connection with the slide bushing **1231**, where the holding bolt **1232** abuts against the connecting post **132**; a plurality of catching grooves **1211** are provided at the upper side of the inner plate **121**; a second gas channel **1321** is provided inside the connecting post **132**, the upper end of the second gas channel **1321** communicating with the catching grooves **1211**; a second cavity **1311** is provided at the inner side of the output seat **131**, a third piston **1312** being slidably attached inside the second cavity **1311**, where the third piston **1312** partitions the second cavity **1311** into an upper chamber disposed at the upper side of the third piston **1312** and a lower chamber disposed at a lower side of the third piston **1312**, the upper chamber communicating with the second gas channel **1321**, the lower chamber communicating with the air; the lower chamber is provided with a spring **1313** exerting an upward force against the third piston **1312**; a connecting rod **1314** is securely connected to the upper side of the third piston **1312**, and the upper end of the connecting rod **1314** abuts against the slide bushing **1231**.

(36) With the above arrangement, after the inner cylinder is completely worked and the squeezing roll **19** compresses the baseplate **22** tightly against the inner plate **121**, it is easy to pull the mandrel **14** out of the inner cylinder; to work the outer cylinder, the outer plate **122** moves downward to bring the third piston **1312** to move downward, where the third piston **1312** draws out the air in the catching grooves **1211** such that the catching grooves **1211** securely hold the blank **21**; after the outer cylinder is completely worked, the first piston rod **153** moves downward to increase the air pressure in the squeezing roll **19**, whereby the squeezing roll **19** is easily pulled out from the outer cylinder, which improves demoulding efficiency of the disclosure.

(37) Specifically, after the cylindrical structure is completely worked by the existing splitting spinning apparatus, the demoulding is strenuous since the inner cylinder is compressed tightly

against the mandrel **14**. In contrast, demoulding is an easy job in the disclosure; as illustrated in FIG. **4**, after the roller **16** finishes working of the inner cylinder, it continuously works the outer cylinder, as illustrated in FIG. **7**, the squeezing roll **19** first compresses the baseplate **22** tightly against the inner plate **121**; at this point, the endplate **192**, the squeezing roll **19**, the blank **21**, and the first piston rod **153** form a sealed cavity; at which time, the first drive mechanism drives the mandrel **14** to move upward, whereby the mandrel **14** is extracted out of the inner cylinder and the ambient air enters the sealed cavity via the first gas channel **1921** and the one-way valve **1922**; at this point, the air pressures inside and outside the squeezing roll **19** are substantially consistent; after the holding bolt **1232** is unscrewed, the slide bushing **1231** slides along the connecting post **132** till the lower end of the connecting post **132** and then the holding bolt **1232** is screwed tightly to thereby hold the slide bushing **1231** to the lower end of the connecting post **132**; at which time, the outer plate **122** is disposed below the inner plate **121**, and under the action of the slide bushing **1231**, the connecting rod **1314** and the third piston **1312** move downward to compress the spring **1313**; when the third piston **1312** moves downward, the volume of the upper chamber is enlarged due to extraction of the air out of the catching grooves **1211**, whereby the cylindrical structure is caught by the catching grooves **1211**; after working of the outer cylinder is complete, as illustrated in FIG. **8**, the first piston rod **153** moves downward to get close to the inner cylinder, which, on one hand, lowers the height of the mandrel **14** to facilitate next-pass processing, and on the other hand, shrinks the volume inside the sealed cavity and increases the pressure in the sealed cavity such that the air in the sealed cavity exerts a downward force to the cylindrical structure; when the squeezing roll **19** moves upward, it is pulled out from the outer cylinder, and the cylindrical structure is caught to the catching grooves **1211**, as illustrated in FIG. **9**; finally, the holding bolt **1232** is unscrewed such that the slide bushing **1231** slides upward, causing the upper side of the outer plate **122** to be in flush with the upper side of the inner plate **121**; under the action of the spring **1313**, the third piston **1312** is reset, and air re-enters the catching grooves **1211**, whereby the catching grooves **1211** release the cylindrical structure, it is convenient for the operator to remove the cylindrical structure from the catching grooves **1211**.

(38) It is understood that modifications and alternations may be made by those skilled in the art according to the description above, and all such modifications and alternations shall fall within the protection scope of the appended claims.

Claims

1. A splitting spinning apparatus, comprising a base, a turntable for supporting a blank, a rotating mechanism for driving the turntable, a mandrel disposed above the turntable, a first drive mechanism for driving the mandrel to move along a vertical direction, a roller provided at one side of the turntable, a second drive mechanism for driving the roller to move along a radial direction of the blank, a third drive mechanism for driving the roller to move along the vertical direction and disposed on the base, a squeezing roll disposed above the turntable, and a fourth drive mechanism for driving the squeezing roll to move along the vertical direction, wherein the squeezing roll and the mandrel are coaxially arranged, the turntable comprises an inner plate rotatably connected to the base, an outer plate sleeved outside of the inner plate and slidable along an axial direction of the inner plate, and a hold mechanism for holding the outer plate in place, the blank has a diameter greater than a diameter of the inner plate and less than an outer diameter of the outer plate, an upper side of the inner plate is flush with an upper side of the outer plate, an outer diameter of the squeezing roll is equal to the diameter of the inner plate, the roller comprises a first state and a second state, when the roller is in the first state, one end of the roller proximal to the turntable tilts downward for the roller to splitting spin an upper side of the blank to form an inner cylinder, when the roller is in the second state, one end of the roller proximal to the turntable tilts upward, such that the roller spins a baseplate from the blank to form an outer cylinder, a lifting platform is

disposed at one side of the turntable, the third drive mechanism is disposed on the base and connected to the lifting platform, an upper side of the lifting platform is slidably connected to a sliding seat, the second drive mechanism is disposed on the lifting platform and connected to the sliding seat, a roller carrier is provided at one side of the sliding seat proximal to the turntable, one end of the roller carrier proximal to the turntable is rotatably connected to the roller, the roller carrier is rotatably connected to the sliding seat, whereby state switching of the roller is enabled by turning the roller carrier, and the sliding seat is provided with a locking mechanism for locking the roller carrier.

2. The splitting spinning apparatus according to claim 1, wherein the roller carrier comprises a rotating seat rotatably connected to the sliding seat and a bracket securely connected to one side of the rotating seat proximal to the turntable; the roller is rotatably connected to one end of the bracket proximal to the turntable; the locking mechanism comprises a retaining ring securely connected to the sliding seat and a locking bolt thread-fitted to the retaining ring, the rotating seat is rotatably connected inside the retaining ring, and the locking bolt abuts against the rotating seat.

3. The splitting spinning apparatus according to claim 2, wherein a first limiting rod is securely connected to an upper side of the retaining ring, a second limiting rod is securely connected to a lower side of the retaining ring, a rotating rod is securely connected to the rotating seat, when the rotating rod abuts against the first limiting rod, the roller is disposed in the first state, and when the rotating rod abuts against the second limiting rod, the roller is disposed in the second state.

4. A splitting spinning apparatus, comprising a base, a turntable for supporting a blank, a rotating mechanism for driving the turntable, a mandrel disposed above the turntable, a first drive mechanism for driving the mandrel to move along a vertical direction, a roller provided at one side of the turntable, a second drive mechanism for driving the roller to move along a radial direction of the blank, a third drive mechanism for driving the roller to move along the vertical direction and disposed on the base, a squeezing roll disposed above the turntable, and a fourth drive mechanism for driving the squeezing roll to move along the vertical direction, wherein the squeezing roll and the mandrel are coaxially arranged, the turntable comprises an inner plate rotatably connected to the base, an outer plate sleeved outside of the inner plate and slidable along an axial direction of the inner plate, and a hold mechanism for holding the outer plate in place, the blank has a diameter greater than a diameter of the inner plate and less than an outer diameter of the outer plate, an upper side of the inner plate is flush with an upper side of the outer plate, an outer diameter of the squeezing roll is equal to the diameter of the inner plate, a top plate is disposed above the base, the top plate being securely connected to the base via a connecting plate, the first drive mechanism is configured as a first oil cylinder, the first oil cylinder comprising a first cylinder body securely connected to a lower side of the top plate, a first piston slidably connected inside the first cylinder body, a first piston rod securely connected to a lower side of the first piston, and a first oil pump connected to the first cylinder body, and a lower end of the first piston rod is rotatably connected to the squeezing roll.

5. The splitting spinning apparatus according to claim 4, wherein the fourth drive mechanism comprises a shell sleeved over the first cylinder body, a first cavity surrounding the first cylinder body is formed between the shell and the first cylinder body, the fourth drive mechanism further comprises a second piston slidably connected inside the first cavity, a second piston rod securely connected to a lower side of the second piston, and a second oil pump in communication with the first cavity, a mounting plate is securely connected to a lower end of the second piston rod, and the squeezing roll is rotatably connected to a lower side of the mounting plate.

6. The splitting spinning apparatus according to claim 5, wherein a rotating plate is disposed between the mounting plate and the squeezing roll, the rotating plate and the mounting plate are both of an annular shape, the first piston rod passes through the rotating plate and the mounting plate, the rotating plate and the squeezing roll are detachably connected by a guide ring extends upward and is securely connected to an inner side of the rotating plate, a limiting protrusion extends outward and

is securely connected to an upper side of the guide ring, an inner side of the mounting plate is disposed between the limiting protrusion and the rotating plate, a rolling bearing is disposed between the mounting plate and the guide ring, a thrust bearing is disposed between the limiting protrusion and the mounting plate, and the thrust bearing is further disposed between the mounting plate and the rotating plate.

7. The splitting spinning apparatus according to claim 6, wherein the rotating mechanism comprises an output seat rotatably connected to the base, the inner plate is disposed at an upper side of the output seat and securely connected to the output seat via a connecting post, the rotating mechanism further comprises a driven wheel securely coupled to the output seat, an electric motor disposed a lower side of the base, and a driving wheel attached to the electric motor, the driving wheel is in transmission connection with the driven wheel via a belt, and a radius of the driving wheel is smaller than a radius of the driven wheel.

8. The splitting spinning apparatus according to claim 7, wherein an endplate is securely attached to an upper end of the squeezing roll, the endplate is detachably connected to the rotating plate, the first piston rod passes through the endplate and is in sealed connection with the endplate, an O-shaped ring is disposed between the first piston rod and the endplate, a first gas channel passing through upper and lower sides of the endplate is provided on the endplate, a lower end of the first gas channel extends until an inner side of the squeezing roll, the first gas channel is provided with a one-way valve facing the inner side of the squeezing roll, the hold mechanism comprises a slide bushing sleeved over the connecting post and a holding bolt in threaded connection with the slide bushing, the holding bolt abuts against the connecting post, a plurality of catching grooves are provided at an upper side of the inner plate, a second gas channel is provided inside the connecting post, an upper end of the second gas channel communicates with the catching grooves, a second cavity is provided at an inner side of the output seat, a third piston is slidably attached inside the second cavity, the third piston partitions the second cavity into an upper chamber disposed at an upper side of the third piston and a lower chamber disposed at a lower side of the third piston, the upper chamber communicates with the second gas channel, the lower chamber communicates with air, the lower chamber is provided with a spring exerting an upward force against the third piston, a connecting rod is securely connected to an upper side of the third piston, and an upper end of the connecting rod abuts against the slide bushing.
